9A Lesson 1

Environmental variation

Year 9 Science · 9A Genetics and evolution · Student Book pages 5 to 7

Name _____ Class _____ Date _____

Today I can:

☐ I can describe abiotic and biotic environmental factors and give examples of environmental variation.

☐ I can tell continuous variation from discontinuous variation, and explain why environmental variation can make classification difficult.

Key words

species a group of organisms that can breed together to produce offspring that can also breed

extinct a species with no living members left

environment the surroundings of an organism

abiotic factor a physical (non-living) part of the environment, e.g. temperature

biotic factor the activities of other organisms, e.g. predation, disease

characteristic a feature of an organism

variation the differences between the characteristics of organisms

environmental variation variation caused by an organism's environment

continuous variation variation that can have any value between two points

discontinuous variation variation with only a limited set of possible values

classification sorting organisms into groups

Core Everyone does this section.

1. Match each key word to its meaning. Draw a line.

species	the surroundings of an organism
extinct	a non-living factor, e.g. temperature or light
environment	a species with no living members left
abiotic factor	a group of organisms that can breed together to produce offspring that can also breed
biotic factor	the activities of other organisms, e.g. predation or disease

2. Tick one box in each row to show whether the factor is abiotic or biotic.

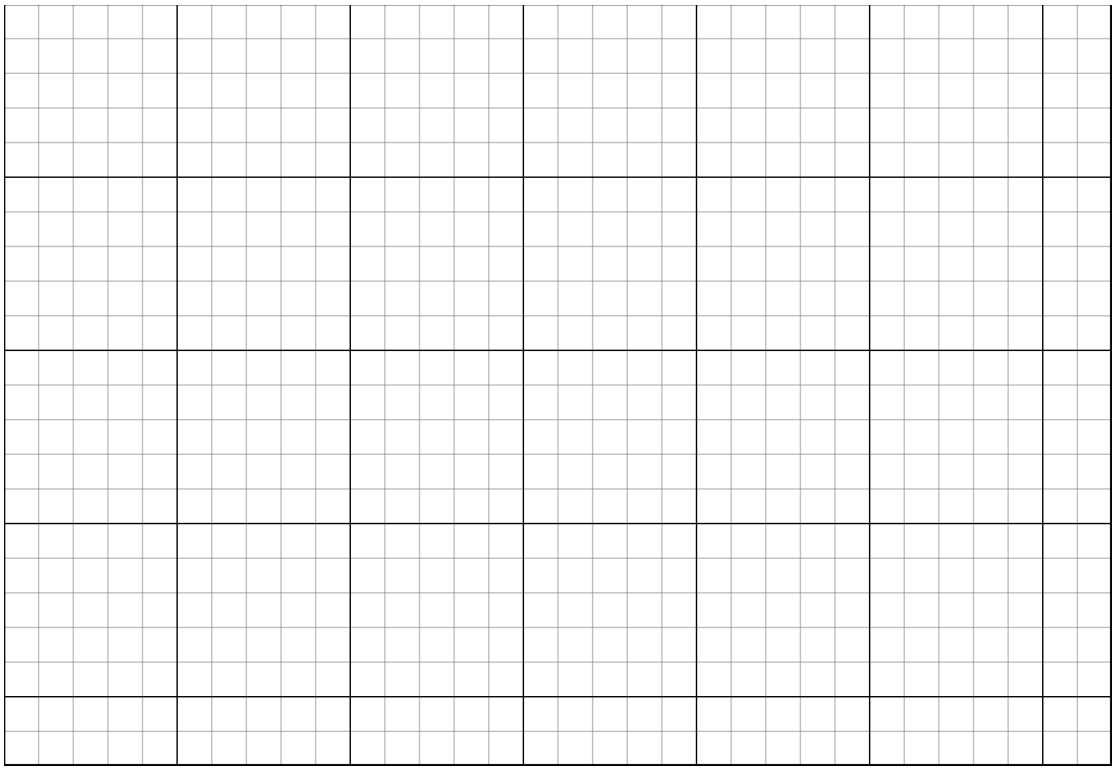
Environmental factor	Abiotic	Biotic
temperature		
predation (being hunted)		
amount of light		
competition for food		
rainfall		
infectious disease		

3. Twenty potatoes of the same variety were weighed. Their masses in grams were: 142, 118, 165, 151, 137, 174, 129, 158, 146, 183, 121, 160, 149, 133, 171, 155, 140, 127, 163, 148.

a) Complete the tally chart.

Mass range (g)	Tally	Number of potatoes
110 to 129		
130 to 149		
150 to 169		
170 to 189		

b) Plot a bar chart of the number of potatoes in each mass range on the grid below. Label both axes.



c) Is potato mass continuous or discontinuous variation? Give a reason.

●● Stretch When Core is finished.

S1. Describe two examples of environmental variation in humans and state the environmental factor that causes each one.

Example of environmental variation	Environmental factor that causes it

S2. Tick one box in each row to show whether the variation is continuous or discontinuous.

Characteristic	Continuous	Discontinuous
height		
blood group		
hand span		
number of fingers		
length of a leaf		
able to roll the tongue (yes or no)		

S3. The twenty potatoes in question 3 were all the same variety. Explain why they still varied in mass.

●●● Challenge Explain in full sentences.

C1. A plant on a seashore has pink flowers and is 10 cm tall. On a nearby island, similar plants are over 50 cm tall with dark purple flowers. Scientists grew seeds from both places in the same laboratory. All the plants grew to look the same.

a) What does this tell you about the plants from the two areas?

b) Use this example to explain how environmental variation can make it difficult to identify plant species.

C2. In 2003, bones of adult humans just over 1 m tall were found on Flores Island. Some scientists think they were our species but had too little iodine in their diet; others think they were a different species that is now extinct. Suggest what evidence would help scientists decide, and explain your reasoning.

My thinking (any language)

A classmate says: 'All variation is caused by the environment.' Do you agree? Explain why, or give an example that shows they are wrong.

Word bank abiotic factor · biotic factor · environment · environmental variation · continuous variation ·

discontinuous variation · species · classification

✂ ---------- cut here

Exit ticket 9A Lesson 1: Environmental variation Name _____

1. Give one abiotic factor and one biotic factor in a pond.

_____

2. Is a scar an example of inherited or environmental variation?

_____

3. Is foot length continuous or discontinuous variation? Give a reason.

_____

Today I can: Goal 1 ☐ Goal 2 ☐